

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Electrical and Computer Engineering SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTECPE502B - CONVENTIONAL AND RENEWABLE ENERGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The main function of a thermal power plant is to:
A. Convert electrical energy into mechanical energy
B. Convert chemical energy of fuel into electrical energy
C. Convert solar energy into electrical energy
D. Store electrical energy
- 2 The function of a boiler in a thermal power plant is to:
A. Condense steam
B. Heat feed water into steam
C. Generate electricity
D. Cool exhaust gases
- 3 The function of a condenser is to:
A. Increase steam pressure
B. Convert steam into water
C. Superheat steam
D. Pump water
- 4 Which turbine is used for high head and low discharge?
A. Kaplan turbine
B. Francis turbine
C. Pelton turbine
D. Propeller turbine
- 5 The function of a penstock is to:
A. Store water
B. Carry water from reservoir to turbine
C. Increase water pressure
D. Control turbine speed
- 6 The surge tank in a hydro power plant is used to:
A. Store excess water
B. Reduce water hammer effect
C. Increase head of water
D. Control generator output

- 7 The main nuclear fuel used in nuclear power plants is:
 - A. Uranium-238
 - B. Uranium-235
 - C. Thorium-232
 - D. Plutonium-240
- 8 The function of a moderator in a nuclear reactor is to:
 - A. Absorb neutrons
 - B. Increase neutron speed
 - C. Slow down fast neutrons
 - D. Stop nuclear reaction
- 9 The prime mover used in a diesel power plant is:
 - A. Steam turbine
 - B. Gas turbine
 - C. Diesel engine
 - D. Hydraulic turbine
- 10 The main function of a compressor in a gas power plant is to:
 - A. Compress fuel
 - B. Compress air before combustion
 - C. Cool exhaust gases
 - D. Increase turbine speed
- 11 Which device converts solar energy directly into electrical energy?
 - A. Solar thermal collector
 - B. Solar inverter
 - C. Solar photovoltaic cell
 - D. Battery
- 12 The function of a gearbox in a wind turbine is to:
 - A. Increase torque
 - B. Reduce wind speed
 - C. Increase rotational speed
 - D. Store mechanical energy

Q2. Solve the following.

- A) What factors are taken into account while selecting the site for a steam power plant? 6
- B) Explain boiler and superheater in steam generation. 6

Q3. Solve the following.

- A) Explain the working of a Pelton turbine with the help of a neat diagram. 6
- B) Define and explain the terms 6
 1. Connected load 2. Maximum demand
 3. Demand factor 4. Average load.

Q4. Solve Any Two of the following.

- A) Discuss the advantages of interconnected grid system. 6
- B) Explain the terms load factor and diversity factor. How do these factors influence the cost of generation? 6
- C) State the advantages and disadvantages of nuclear power plant 6

Q5. Solve Any Two of the following.

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| A) | Explain with a neat sketch the layout of a gas turbine power plant. | 6 |
| B) | Explain working of a single basin tidal power plant. | 6 |
| C) | Explain the working of biomass plant. | 6 |
- Q6. Solve Any Two of the following.
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|----|--|---|
| A) | Explain the working of geothermal plant | 6 |
| B) | Explain with diagram a wind electric generating plant. | 6 |
| C) | Discuss the advantages of interconnected grid system. | 6 |

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